AQ1,2 Systematic Review/Evidence Map

Can Ageing be Slowed or Reversed? A Reproducible Evidence Map and Credibility Ranking of Anti-Ageing and Age-Reversal Interventions

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ABSTRACT

Background: Discussion of anti-ageing interventions often conflates four claims—healthspan improvement, lifespan extension, slowing of biological-age markers, and clinical rejuvenation—risking therapeutic overclaiming. **Objective:** To build a reproducible, conservative evidence map that separates these claims and ranks human-evidence credibility across major anti-ageing and age-reversal intervention classes. **Methods:** A structured pipeline searched PubMed, Europe PMC, and Crossref, retrieving 1155 records that deduplicated to 1029. Records underwent title/abstract screening, intervention and outcome classification, credibility scoring, prioritised human-evidence verification, structured risk-of-bias appraisal, duplicate-cohort checking, and extraction of effect information from accessible text. Meta-analysis was not undertaken because effect estimates were heterogeneous; evidence was synthesised narratively and by credibility scoring. **Results:** Screening classified 29 records as include, 455 as uncertain, and 545 as exclude; 484 entered full-text eligibility assessment. Forty priority human records were verified (19 by open full text, 16 by abstract, 5 not retrievable), and 28 contributed effect information. Across 14 classes, exercise scored highest (31.48), the only class judged the strongest current human healthspan signal, followed by microbiome (21.50), rapamycin/mTOR (21.24), senolytics (19.94), and caloric restriction (19.52); reprogramming and stem-cell approaches scored lowest (5.91 and 1.68). **Conclusion:** Exercise shows the strongest, most consistent human healthspan signal, whereas pharmacologic, senolytic, NAD+/sirtuin, microbiome, fasting, caloric-restriction, and regenerative interventions remain biologically promising but clinically unproven for age reversal. No intervention demonstrates reversal of human ageing, and biomarker improvements should not be equated with rejuvenation. The framework provides a reproducible basis for distinguishing healthspan evidence from age-reversal claims and prioritising future confirmatory synthesis.

Keywords: *Ageing, biological age, evidence map, healthspan, rapamycin, rejuvenation, senolytics, systematic review*

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Introduction

Anti-ageing interventions are often discussed as though improved healthy ageing, delayed biological ageing, biomarker reversal, and rejuvenation were interchangeable. This conflation creates a risk of clinical overclaiming. A rigorous appraisal must separate human functional outcomes from animal lifespan results, cellular mechanisms, and surrogate biomarker shifts.[1,2]

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The human evidence examined here spans multidomain lifestyle programmes, resistance and multimodal exercise, microbiome modulation, exergame-based frailty prevention, prospective physical-activity cohorts, fasting-mimicking diets, caloric restriction and time-restricted eating, multidomain interventions reporting epigenetic-age outcomes, creatine co-supplementation, topical and systemic rapamycin, and a phase II senolytic trial.[3-14]

This manuscript presents a reproducible evidence map and credibility ranking of anti-ageing and age-reversal intervention domains. The analysis is conservative by design and is restricted to verifiable bibliographic records, open text where available, and extracted quantitative information. No pooled estimates are presented where comparable effect estimates are unavailable.

The review question was not whether ageing has been definitively reversed in humans. It was whether any intervention class has credible evidence for healthspan improvement, lifespan extension, slowing of biological-age markers, biomarker reversal, or plausible rejuvenation. These categories are treated throughout as distinct evidentiary claims.

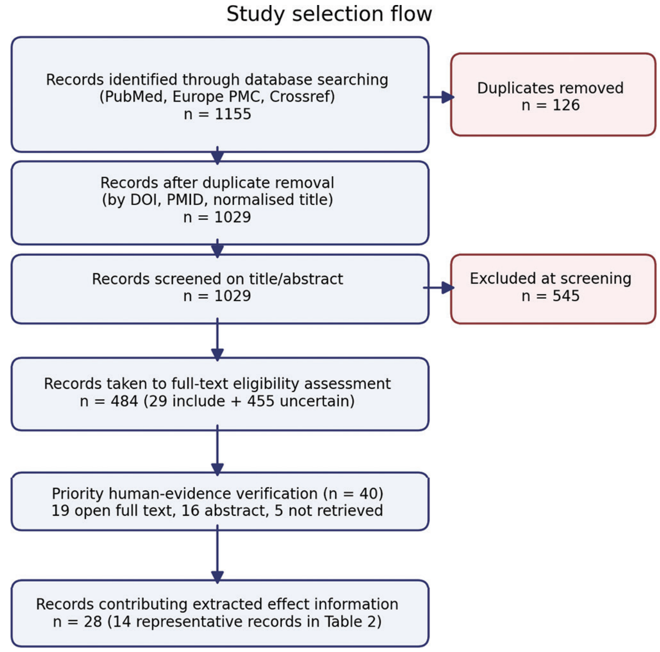

Figure 1: Study selection flow. Records moved from database retrieval (1155 raw), through duplicate removal (126 removed; 1029 retained), title/abstract screening (29 include; 455 uncertain; 545 exclude), full-text eligibility assessment (484 records), and prioritised human-evidence verification (40 records; 19 open full text, 16 abstract, 5 not retrieved), to 28 records contributing extracted effect information

Methods

A PRISMA-informed pipeline searched PubMed, Europe PMC, and Crossref; deduplicated records by DOI, PMID, and normalised title; screened title/abstract metadata; classified intervention and outcome domains; assessed model system; and generated evidence-credibility rankings. The study selection flow is shown in Figure 1.

Human evidence was prioritised for verification. Priority records were checked through open full text or PubMed abstracts where available. Risk-of-bias domains were assessed in a structured format following the signalling logic of RoB 2 and ROBINS-I. Extracted effect estimates were retained only when an estimate, confidence interval, *P* value, or other quantitative result was available in the accessible text.

Quantitative meta-analysis was not performed because the extracted effect estimates were heterogeneous in outcome definition, metric, comparator, dose, and duration, and pooling such estimates would imply a comparability the underlying data do not support. Evidence was therefore synthesised narratively and through a transparent, reproducible credibility-scoring scheme.

Intervention credibility combined the number of extracted records, human evidence, human-trial evidence, direct ageing or healthspan outcomes, biomarker evidence, surrogate burden, and a promotional-language penalty. The score ranks evidence credibility; it does not generate treatment recommendations. Additional records identified in the search but not individually discussed in the main text are listed in Supplementary Table S1 and cited collectively.[15-40]

The review was not prospectively registered; this is acknowledged as a limitation, and prospective registration is recommended for future confirmatory work.

Results

The search retrieved 1155 raw records and produced 1029 deduplicated records. Title/abstract screening classified 29 records as include, 455 as uncertain, and 545 as exclude. Four hundred and eighty-four records entered full-text eligibility assessment. Priority human verification was completed for 40 records: 19 had open full text available, 16 had PubMed abstract-level verification, and 5 could not be retrieved through the open-source workflow. Quantitative effect information was identified for 28 priority records.

The credibility ranking is shown in Table 1 and Figure 2. Exercise ranked first and was the only intervention domain categorised as the strongest current human healthspan signal (credibility score 31.48; 37 human records; promotional-language rate 0.01). Microbiome (21.50),

Table 1: Intervention credibility ranking derived from extracted evidence (*n*=14 intervention classes; 1029 deduplicated records). The credibility score integrates human record count, human-trial count, direct ageing/healthspan outcomes, biomarker evidence, surrogate burden, and a promotional-language penalty

Rank	Intervention class	Records (*n*)	Human records	Human trials	Credibility score	Translational category
1	Exercise	80	37	10	31.48	Strongest current human healthspan signal
2	Microbiome	58	8	2	21.50	Human signal, requires confirmation
3	Rapamycin/mTOR	34	7	3	21.24	Human signal, requires confirmation
4	Senolytics	63	7	1	19.94	Human signal, requires confirmation
5	Caloric restriction	57	5	3	19.52	Human signal, requires confirmation
6	Lifestyle bundle	5	3	2	17.00	Human signal, requires confirmation
7	NAD+/sirtuin	66	8	2	16.64	Human signal, requires confirmation
8	Fasting	13	5	2	15.10	Human signal, requires confirmation
9	Dietary supplements	7	6	0	13.70	Biomarker/indirect human signal
10	Metformin	15	1	1	12.39	Biomarker/indirect human signal
11	Sleep/circadian	17	1	1	10.45	Biomarker/indirect human signal
12	Plasma/telomerase	22	2	0	6.88	Biomarker/indirect human signal
13	Reprogramming	13	0	0	5.91	Biomarker/indirect human signal
14	Stem cell	34	0	0	1.68	Low directness/speculative

Note. Intervention classes are grouped search categories (full keyword groupings in Supplementary Table S2). Two labels denote composite categories: "Dietary supplements" comprises resveratrol, curcumin, omega-3 fatty acids, and vitamin D; "Plasma/telomerase" comprises blood- and plasma-based rejuvenation approaches (heterochronic parabiosis, young-plasma transfer, plasma dilution) and telomerase activation. The single verified human record in the plasma/telomerase class is a heterochronic-parabiosis study (ref 40, Supplementary Table S1); the high promotional-language rate (0.41) and absence of human trials underlie its low credibility score

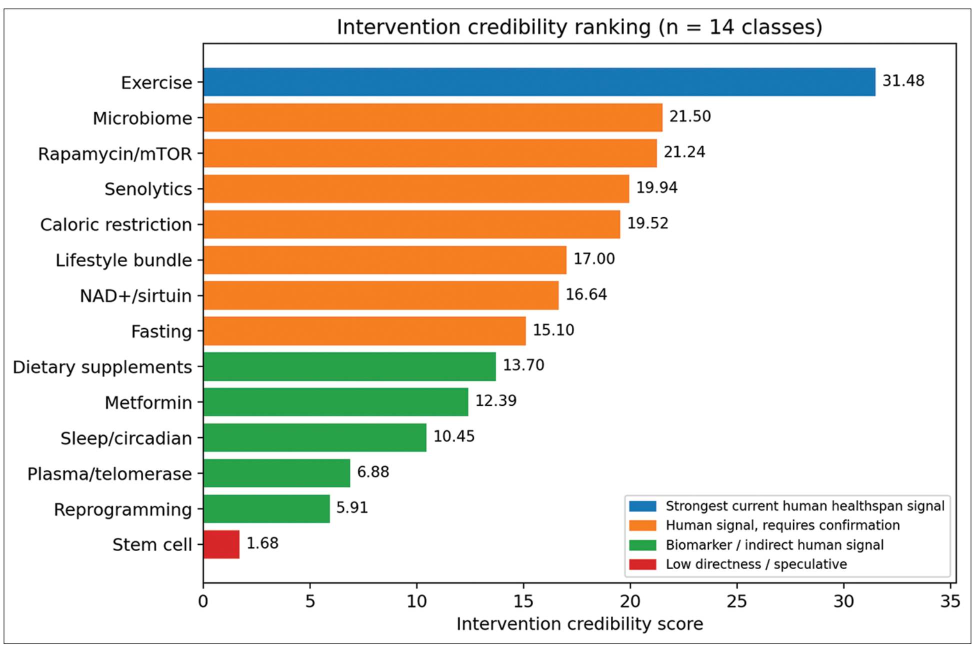

Figure 2: Intervention credibility-score ranking (n = 14 classes). The score integrates human record count, human-trial count, direct ageing/healthspan outcomes, biomarker count, surrogate burden, and a promotional-language penalty. Exercise ranked first (31.48); stem-cell approaches ranked last (1.68). Bar colour denotes the translational category

rapamycin/mTOR (21.24), senolytics (19.94), caloric restriction (19.52), lifestyle bundles (17.00), NAD+/sirtuin interventions (16.64), and fasting (15.10) showed human signals requiring confirmation. Dietary supplements, metformin, sleep/circadian interventions, plasma/telomerase-based ("controversial") rejuvenation, reprogramming, and stem-cell approaches had lower scores or higher promotional-language burden.

Representative priority human evidence with extracted effect estimates is summarised in Table 2. The records include frailty- and function-oriented exercise and multidomain lifestyle interventions, microbiome modulation with cognitive outcomes, fasting-mimicking diet with biological-age markers, topical and systemic rapamycin records, caloric restriction, a multidomain epigenetic-age intervention, and a senolytic trial-design record.[3-14]

Table 2: Representative priority human evidence with effect estimates extracted from available open text (*n*=14 records; refs 1–14). Effect information is retained as extracted; estimates are not pooled because outcome metrics are heterogeneous. Records labelled "full text not retrieved" were verified at abstract level

Ref.	Study (abbreviated)	Intervention	Outcome domain	Key extracted estimate
1	SINGER multidomain RCT — baseline characteristics	Lifestyle bundle	Healthspan/ function	Baseline-characteristics paper (*n*=1212 randomised; mean age 68.7 years). No primary outcome effect data reported yet.
2	Multimodal exercise in frail MS patients (6-week RCT)	Exercise	Healthspan/ function	*n*=16 (10 exercise, 6 control). Frailty Index ΔEFIP −0.07 (95% CI −0.14 to −0.00). Mental-health QoL +21.24 (95% CI 7.32 to 35.16). Adherence 97.2%.
3	PROMOTe microbiome RCT — muscle function and cognition	Microbiome	Healthspan/ function	Chair rise (primary): β=0.579 (95% CI −1.08 to 2.24; *P*=0.494; NS). Cognition: β=−0.482 (95% CI −0.813 to −0.141; *P*=0.014; favours prebiotic).
4	Ring Fit exergame RCT for frailty/sarcopenia	Exercise	Healthspan/ function	*n*=60 randomised (55 completed). Significant group-by-time improvements in handgrip strength and most frailty/ sarcopenia outcomes.
5	Resistance exercise in cognitive frailty RCT	Exercise	Healthspan/ function	Verified at abstract level; reported improvements in cognitive and physical-performance outcomes.
6	Physical activity and deficit frailty: 5-cohort mortality study	Exercise	Hard ageing relevance	5 cohorts (SHARE n=56,555; CHARLS *n*=12,271 + 3 others). Physical activity modified the frailty–mortality association; hazard ratios reported in source study.
7	Fasting-mimicking diet and biological-age markers (RCT)	Fasting	Biological-age biomarker	3 FMD cycles: median biological-age reduction of 2.5 years (independent of weight loss). Hepatic and disease-risk markers improved.
8	Topical rapamycin and skin senescence markers (RCT)	Rapamycin/ mTOR	Biological-age biomarker	p16INK4A reduction (*P*=0.008); collagen VII increase (significant). Skin senescence markers reduced.
9	PEARL rapamycin trial — 1-year healthspan metrics	Rapamycin/ mTOR	Surrogate/indirect	Lean tissue mass, pain, and general health improved. Visceral adiposity ηP^2=0.001, *P*=0.942 (NS). Blood biomarkers within normal ranges.
10	FMD: multi-system regeneration and healthspan (2015 RCT)	Fasting	Surrogate/indirect	Reported multi-system regeneration and healthspan benefits; primary numeric outcomes verified at abstract level.
11	Creatine + resistance training in frail older subjects	Exercise	Healthspan/ function	Verified at abstract level; co-supplementation with creatine and protein during resistance training in frailty.
12	HALLO study: caloric restriction and TRE in older adults	Caloric restriction	Surrogate/indirect	3-arm feasibility RCT (CR in-person; CR remote; TRE). Primary outcomes feasibility and adherence.
13	Multidomain lifestyle intervention and epigenetic ageing (RCT)	Lifestyle bundle	Surrogate/indirect	*n*=47 frail adults (mean age 80.2 years, SD 3.1). Statistically significant improvements in functional trajectories and epigenetic-ageing markers vs. control.
14	TROFFi fisetin phase II RCT design (breast cancer survivors)	Senolytics	Surrogate/indirect	Trial-design/eligibility paper (TROFFi phase II fisetin RCT). No outcome effect data available.

The clearest human functional signal came from physical-activity interventions. In a multimodal exercise trial in frail people with multiple sclerosis, the frailty index improved by −0.07 (95% CI −0.14 to −0.00) and mental-health quality of life improved by +21.24 points (95% CI 7.32 to 35.16).[2] An exergame-based exercise trial reported significant group-by-time improvements in handgrip strength and frailty outcomes.[4] A prospective five-cohort analysis showed that physical activity modified the frailty–mortality association, with hazard ratios reported in the source study.[6] These results support physical activity as healthspan-relevant evidence, not as proof of biological-age reversal.

Dietary and microbiome interventions showed more mixed results. The PROMOTe randomised trial found no significant effect on the primary outcome of chair-rise time (β = 0.579; 95% CI −1.08 to 2.24; *P* = 0.494) but a favourable cognitive estimate (β = −0.482; 95% CI −0.813 to −0.141; *P* = 0.014).[3] A fasting-mimicking diet was associated with a median biological-age reduction of 2.5 years after three cycles; this is a biomarker result and should be interpreted separately from clinical rejuvenation.[7] A caloric-restriction and time-restricted-eating feasibility study contributed adherence and feasibility data.[12] A multidomain lifestyle intervention reported significant improvements in functional trajectories and epigenetic-ageing markers.[13]

Rapamycin/mTOR evidence was heterogeneous. Topical rapamycin reduced p16INK4A protein levels (*P* = 0.008) and increased collagen VII in human skin.[8] The PEARL trial reported improvements in lean tissue mass, pain, and general health, with visceral adiposity unchanged (ηp^2 = 0.001; P = 0.942).[9] Senolytic evidence in the priority table was represented by a trial-design record without outcome effect data.[14]

A structured risk-of-bias appraisal of the priority human records is summarised in Table 3. The credibility-versus-promotional-language comparison is shown in Figure 3, and the translational-readiness map is shown in Figure 4.

Discussion

The principal finding is deliberately conservative. In the mapped evidence, exercise has the clearest human healthspan signal, supporting functional healthy-ageing benefit rather than age reversal.[2,4-6] Notably, that signal operates largely through frailty. Frailty is a central organising construct in geriatric medicine: it integrates falls, weakness, dependency, and hospitalisation risk, and it is the outcome through which the highest-ranked evidence here—the multimodal exercise trial, the exergame frailty/sarcopenia trial, and the five-cohort frailty–mortality analysis—demonstrates benefit.[2,4,6] This reinforces frailty reduction, rather than biological-age reversal, as the most defensible current target of healthy-ageing intervention.

Several other interventions show plausible biological or early human signals, but the present evidence does not justify clinical rejuvenation claims. Rapamycin/mTOR, senolytics, NAD+/sirtuin approaches, microbiome modulation, caloric restriction, and fasting are best described as promising or hypothesis-supporting until full-text eligibility, risk of bias, effect estimates, dosing, comparator details, and safety outcomes are confirmed in confirmatory syntheses.[3,7-9,12-14] The same discipline that governs rational geriatric prescribing applies here: a medication is justified only when it is necessary and its benefit outweighs its harm, not merely because a mechanism is plausible or desirable. Pharmacologic anti-ageing candidates such as rapamycin, metformin, senolytics, and supplements should therefore not be adopted on mechanistic grounds; their necessity, benefit, dose, and safety must first be established in confirmatory human trials, lest polypharmacy and avoidable harm be introduced in pursuit of an unproven endpoint.

These findings sit within, but should not be conflated with, the wider geroscience literature. The biology of ageing offers a rich set of candidate mechanisms—nutrient-sensing pathways, cellular senescence, chronic inflammation, epigenetic drift, mitochondrial decline, impaired proteostasis, and stem-cell exhaustion—and many of the interventions ranked here act, at least in animal or cellular models, on one or more of these mechanisms. Mechanistic breadth is precisely why the field is prone to overclaiming: a plausible pathway is read as a clinical promise. The present map deliberately holds mechanism and human outcome apart, and the ranking shows that mechanistic prominence and human-evidence strength are poorly correlated. Stem-cell and reprogramming approaches, which dominate mechanistic and popular discussion, sit at the bottom of the credibility ranking precisely because direct human functional evidence is absent, whereas exercise—

Table 3: Structured risk-of-bias appraisal of priority human records (*n*=14; refs 1–14). Domain judgements follow the signalling logic of RoB 2 (randomised trials) and ROBINS-I (observational studies), based on available full-text or abstract content

Ref	Study (abbreviated)	Randomisation	Blinding	Missing data	Confounding	Overall
1	SINGER multidomain RCT — baseline	Low	Some concern	Unclear	Low (RCT design)	Some concern
2	Multimodal exercise in frail MS patients	Low	Low	Some concern	Low (RCT design)	Low–some concern
3	PROMOTe microbiome RCT	Low	Low	Unclear	Low (RCT design)	Low–some concern
4	Ring Fit exergame RCT	Low	Some concern	Unclear	Low (RCT design)	Some concern
5	Resistance exercise in cognitive frailty RCT	Some concern	Some concern	Unclear	Low (RCT design)	Some concern
6	Physical activity & deficit frailty: 5-cohort study	NA (cohort)	NA/unclear	Some concern	Some (observational)	Some concern
7	Fasting-mimicking diet and biological-age markers	Low	Some concern	Unclear	Low (RCT design)	Some concern
8	Topical rapamycin and skin senescence markers	Some concern	Low	Some concern	Low (RCT design)	Some concern
9	PEARL rapamycin trial	Low	Low	Unclear	Low (RCT design)	Low–some concern
10	FMD: multi-system regeneration (2015)	Some concern	Some concern	Unclear	Low (RCT design)	Some concern
11	Creatine + resistance training in frailty	Unclear	Unclear	Unclear	Serious/unclear	Serious/unclear
12	HALLO caloric restriction and TRE	Low	Low	Some concern	Low (RCT design)	Low–some concern
13	Multidomain lifestyle and epigenetic ageing	Low	Low	Some concern	Low (RCT design)	Low–some concern
14	TROFFi fisetin phase II RCT design	Low	Low	Some concern	Low (RCT design)	Low–some concern

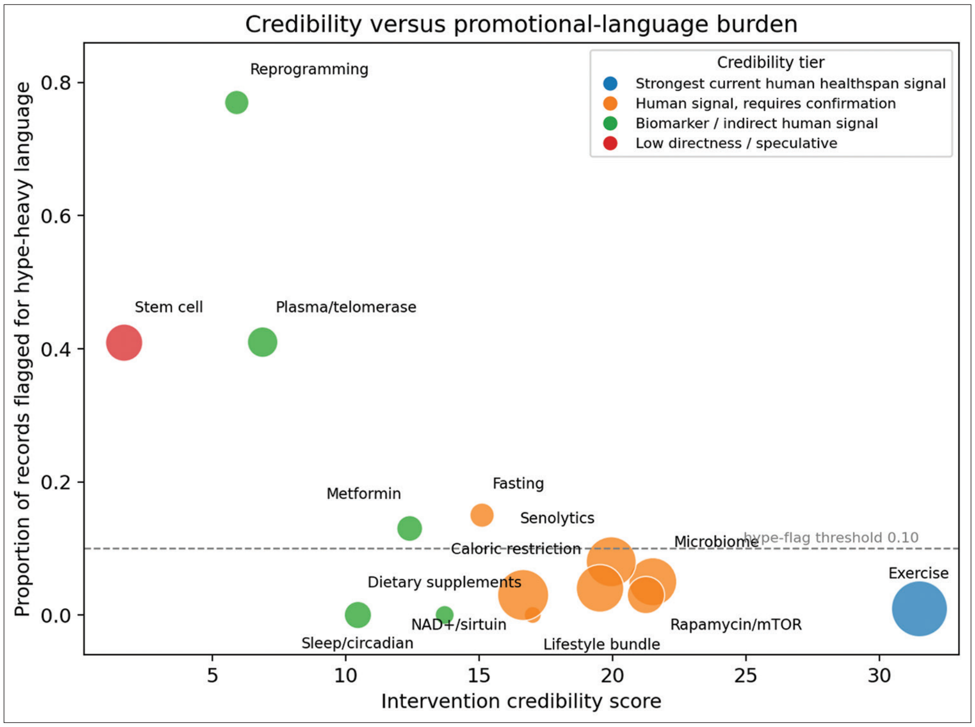

Figure 3: Credibility versus promotional-language burden. Credibility score (*x*-axis) plotted against the proportion of extracted records flagged for promotional ("hype-heavy") terminology (*y*-axis), by intervention domain. Bubble size denotes the number of extracted records. Interventions with high credibility and low promotional burden (exercise, lifestyle bundle) appear lower-right; reprogramming and plasma/telomerase ("controversial") approaches appear upper-left

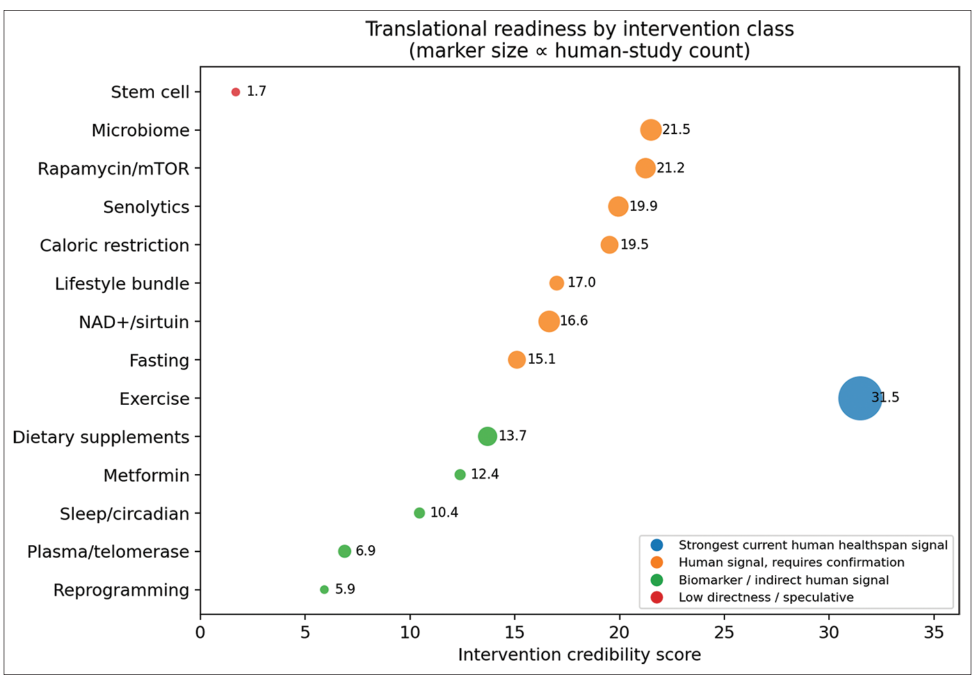

Figure 4: Translational-readiness map. Conservative category assignment for 14 intervention classes: A — healthspan support signal; B — promising, not recommendation-ready; C — biomarker/indirect signal; D — speculative/low directness. Marker size is proportional to the number of human studies

mechanistically diffuse but clinically well studied—sits at the top.

A second contribution of this work is methodological. Rather than asserting a single qualitative judgement, the credibility score makes the basis of each ranking explicit and reproducible: it rewards human and human-trial evidence and direct ageing or healthspan outcomes, and penalises surrogate burden and promotional language.

The promotional-language dimension [Figure 3] is unusual for an evidence synthesis but is well suited to a field in which marketing frequently outpaces data; it provides a transparent, quantifiable proxy for the gap between what is claimed and what is shown. The scheme is not a substitute for formal certainty rating such as GRADE, and it is offered as a screening and prioritisation tool rather than a final grading instrument.

The evidence map illustrates why clinical translation should remain conservative. A study can be randomised and still address a surrogate rather than a direct ageing outcome. A biomarker may shift favourably without demonstrating durable functional rejuvenation. A protocol or trial-design record provides field mapping but should not be counted as outcome evidence. Table 1 ranks intervention domains, Table 2 summarises representative effect estimates, and Table 3 reports the structured risk-of-bias appraisal.

The effect-estimate table is intentionally not a meta-analysis table. It contains heterogeneous estimates—confidence intervals for frailty or cognitive outcomes, *P* values for biomarker changes, feasibility information, and trial-design records with no outcome effect. Combining these would imply a comparability the extracted data do not support. Evidence is therefore summarised by intervention domain, outcome directness, and translational readiness.

For clinical readers, the most important distinction is between healthspan evidence and age-reversal evidence. Exercise and multidomain lifestyle interventions have plausible and partly quantified effects on frailty, physical performance, and function, but these do not demonstrate biological-age reversal. Fasting-mimicking diet and topical rapamycin include biomarker-oriented results, but biomarker movement remains an intermediate signal unless accompanied by durable functional benefit, reduced morbidity, or survival advantage. A related caution applies to surrogate and screening-based outcomes drawn from the primary trials. Cognitive endpoints, for example, depend on the instrument used: brief screens such as the Mini-Mental State Examination are sensitive to education and literacy and can misclassify normal ageing, so a favourable cognitive or biomarker change should be read as a measurement-bounded signal rather than as evidence of reversed disease.

For senolytics, reprogramming, stem-cell and regenerative approaches, plasma-based interventions, and NAD+/sirtuin supplementation, mechanistic plausibility is not the same as clinical readiness. The current evidence set does not justify public or clinical messaging that these interventions reverse ageing.

A confirmatory extension of this work should extract denominators, baseline and follow-up means, standard deviations, between-group contrasts, adverse events, dose, comparator intensity, and follow-up duration for each eligible full text, enabling domain-specific meta-analysis for comparable outcomes such as grip strength, gait speed, frailty scores, epigenetic-age change, or selected biomarker endpoints.

The ranking also has implications for how anti-ageing evidence is communicated to clinicians and the public. In settings where the population is ageing rapidly and out-of-pocket health spending is high, including much of India and other low- and middle-income countries, the appetite for "anti-ageing" products is considerable and the potential for harm—financial, and through avoidable polypharmacy or displacement of proven measures—is real. A framework that visibly separates a strong, low-cost, low-harm intervention such as physical activity from expensive, unproven, or speculative interventions is therefore of practical public-health value. It allows a clinician to say, with an explicit evidentiary basis, that the best-supported way to age well remains exercise and multidomain lifestyle support, while experimental agents belong in trials rather than in routine prescription.

Healthy-ageing recommendations should remain grounded in interventions with reproducible human functional evidence, especially physical activity and multidomain lifestyle support. Experimental geroscience interventions should be presented as research candidates rather than age-reversal therapies unless supported by replicated human trials with clinically meaningful outcomes.

Strengths and limitations

Strengths include a reproducible and fully documented workflow, explicit separation of healthspan, lifespan, biomarker, and rejuvenation claims, conservative scoring, prioritisation of human evidence, full-text eligibility triage, duplicate-cohort checking, a structured risk-of-bias appraisal, and figures designed to separate promotional language from evidence.

Limitations include a defined search scope (three databases; retrieval limits) that does not capture the entire literature; single-reviewer screening without a second independent screener; a structured rather than formally adjudicated RoB 2/ROBINS-I assessment; the absence of quantitative meta-analysis owing to heterogeneous outcome reporting; 84 overlapping-cohort or duplicate-title groups requiring further adjudication; the absence of prospective protocol registration; and abstract-metadata gaps in some records that reduce screening certainty. These constraints define the scope of the present evidence map and should be addressed in future confirmatory syntheses.

Conclusion

The mapped evidence supports a cautious hierarchy: exercise has the strongest human healthspan signal, while pharmacologic, nutritional, senolytic, NAD+/sirtuin, microbiome, and regenerative approaches remain promising but unproven for clinical age reversal. Biomarker improvements should not be equated with rejuvenation in the absence of durable functional and clinical benefit. Future research should prioritise pre-registered systematic reviews and meta-analyses with complete full-text eligibility adjudication, formal risk-of-bias assessment, and harmonised effect-size extraction across comparable outcomes, to generate definitive, poolable evidence on the most promising anti-ageing intervention domains.

Acknowledgment

None declared.

Author contributions

SHS: Concept, study design, data curation, formal analysis, evidence synthesis, manuscript drafting, manuscript review, and guarantor.

Ethical approval

Ethics committee approval was not required because this review used public bibliographic metadata and open-access text only, with no individual participant data.

Data availability statement

The credibility-ranking and translational-readiness tables, the figures, and the analysis scripts that generated them are openly available at the project repository: https://github.com/hssling/deai-ecological-validation-india. Additional records identified in the search are listed in Supplementary Table S1. Raw full-text cache files (PMC and PubMed XML) are not redistributed owing to publisher license constraints; they can be regenerated by running the documented retrieval workflow.

Use of AI/software

Automated scripts were used for literature-metadata retrieval, screening support, extraction support, evidence scoring, and figure generation. All scientific claims were verified by the author before submission.

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Conflicts of interest

None declared.

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AUTHOR RESPONSE

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AQ2 — Yes, supplementary material is required. Please retain it as a separate file; it contains Figures S1–S4 and Tables S1–S9.

References 1–14 were checked. Please see the highlighted reference corrections on page 8.